

MYSQL DATABASE MANAGEMENT SYSTEM

LAB SHEET #5

Title:

- Perform various operators, pattern matching and aggregate functions on MySQL

Objectives:

- To understand and perform various operators like comparison/relational, logical operators and Special operators
- To understand and use aggregate functions on database

Background Concept:

DB Source: https://github.com/datacharmer/test_db

- **Arithmetic operators**

Operators	Description
+	Addition
-	Subtraction
*	Multiplication
/	Division

- **Comparison and Relational operators**

Operators	Description
=	Equal to
>	Greater than
<	Less than
>=	Greater than equal to
<=	Less than equal to
<>	Not equal to

- **Logical operators**

Operators	Description
AND	returns true if both conditions are true
OR	returns true if either condition is true
NOT	returns true if condition is false

- **Special operators**

Operators	Description
IN	checking the value within a set
BETWEEN	checking the value within a range
LIKE	Matching the pattern of characters
IS NULL	checking the null value

- **Pattern Matching**

Wild-card Character	Description
%	matches zero or more characters
_	matches exactly one character

Examples (used with [NOT] LIKE operator)	
'A_n'	matches string with three character that starts with 'A' and ends with 'n' character
'_an_'	matches string with four character that contain substring 'an' between exactly two characters
'Bi%'	matches strings that starts with 'Bi'
%sh%	matches strings that contain substring 'sh' anywhere
____	matches string with exactly four characters
_____%	matches string with at least three character or more

- Aggregate functions

function	Description
MAX (column)	returns the maximum value of a given attribute
MIN (column)	returns the minimum value of a given attribute
AVG (column)	returns the average value of column values
STDDEV (column)	returns standard deviation of column values
SUM (column)	returns the sum of column values
COUNT (column)	returns the total number of rows
VARIANCE (column)	returns the statistical variance

Lab works:

1. Display employee number, employee name, current salary and salary increased by 10% for each employee
2. List employee name and their annual salary
3. Find the name of employee having half month salary greater than 30,000
4. List empno, fullname and hired_date of male employees hired after 1995
5. Display employees who are female or hired before 1990
6. Display employees who are not females.
7. Find employees whose salary is greater than 60,000
8. List employees hired on or before 1992-12-31
9. Find employees working in departments d001 or d002
10. Display employees whose salary is between 50,000 and 80,000
11. Find the employees currently working
12. Find employees whose first name starts with 'A'
13. Display employees whose last name ends with 'son'
14. Find employees whose first name contains 'an'
15. Display employees whose first name has exactly 5 characters
16. Display employees whose last name has at least 5 characters
17. Display employees whose first name doesn't start with 'Ba'
18. Find total number of employees
19. Find average salary of employees
20. Display highest salary of employees
21. Find the lowest salary of employees
22. Find the total amount of salary paid to employees who works in Research department